HYDROGEN PLANT

TEAM -INNOVIBE

Large-Scale Hydrogen Industry Plant from Seawater: Process, Technology, and Economics

Introduction

Hydrogen production, particularly through green methods, is critical to reducing global carbon emissions and fostering a sustainable energy future. One of the most promising approaches is producing hydrogen from seawater, utilizing renewable energy and advanced electrolytic techniques. This large-scale hydrogen industry plant aims to produce **195 tons of green hydrogen per day**, adopting a comprehensive three-stage process: **seawater treatment**, **desalination**, and **hydrogen production through capillary-fed electrolysis.**

The facility will leverage **renewable energy sources**, such as solar and wind, to power the entire operation, ensuring minimal environmental impact. Advanced **water treatment technologies**, including coagulation, flocculation, ultrafiltration, and forward osmosis, will ensure high-quality feedwater for the electrolysis. The **capillary-fed electrolyzers** are designed for high efficiency and durability, targeting over **95% efficiency** in hydrogen production, with thermal, mechanical, and chemical stability being key priorities.

In addition to production, the plant will incorporate **scalable solutions for hydrogen storage**, such as metal hydrides and cryogenic systems, to ensure reliable supply chain management. To further optimize the Levelized Cost of Hydrogen (LCOH), the project will explore innovative strategies, such as enhanced electrolyze designs, local manufacturing of critical components, and integration with **waste heat recovery systems**.

A detailed financial model, encompassing **CAPEX, OPEX**, and lifecycle costs, has been developed to ensure economic viability. Regulatory support from India's growing hydrogen ecosystem, including subsidies, carbon credits, and renewable energy incentives, will be critical to the project's success. The facility will also prioritize **R&D initiatives** to improve process efficiencies and maintain a competitive edge in the global hydrogen market.

This project aligns with India's **National Hydrogen Mission** and positions the country as a leader in the global green hydrogen transition, paving the way for a sustainable and decarbonized future

Sizing of Production Capacity

The hydrogen production capacity of 195 tons per day (TPD) places the plant in the "large-scale" category, serving various industrial applications such as transportation, energy storage, and heavy industry decarbonization. The production capacity was determined based on projected demand growth in the domestic and international markets.

- **Annual Hydrogen Production:**
    - 195 tons/day × 365 days = 7,11,75,000 tons/year.

Given the scale of production, the facility requires a significant amount of seawater, electricity from renewable energy sources (solar and wind), and state-of-the-artelectrolysis.

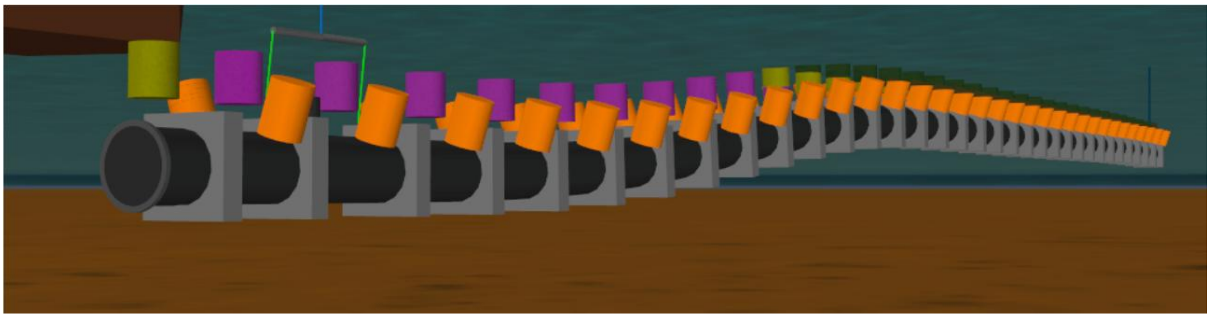
Production Technology

The hydrogen production process can be divided into three main 3 stages:

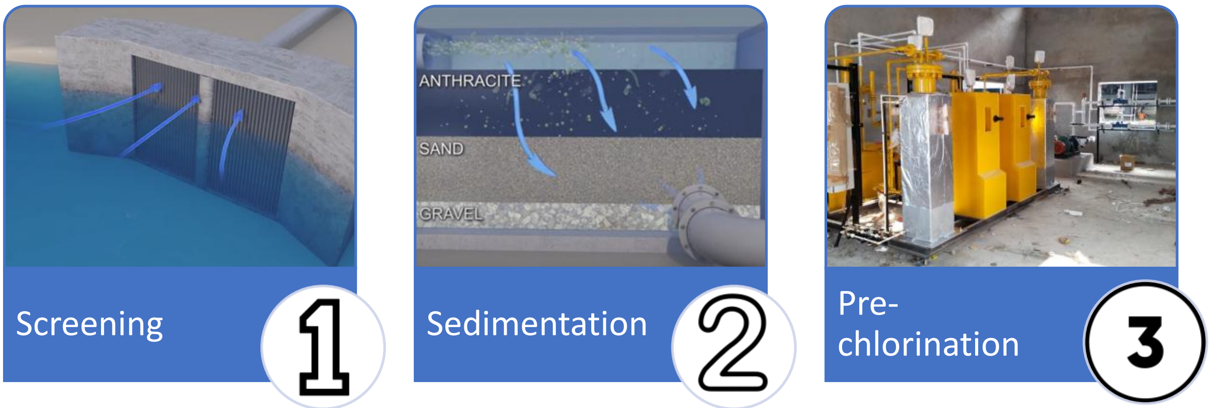
Primary Treatment of Seawater

- **Objective:** To remove large particles, debris, and marine life that could damage equipment or interfere with the subsequent stages of treatment.

  Air-filled Mono Buoyancy Units (MBUs)



- **Process:**
  - **Screening:** Seawater passes through large mesh screens to remove large particles.
  - **Sedimentation:** Gravity-based sedimentation tanks help separate heavier particles.
  - **Pre-chlorination:** Mild chlorination to eliminate microorganisms and prevent biofouling.



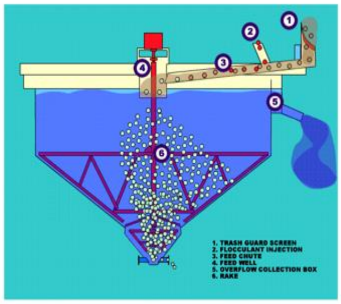
- **Sizing of Equipment:** The plant will be designed to process approximately 30,000 cubic meters of seawater per day. This ensures a steady supply of clean feedstock for the next stage.

Coagulation and Flocculation, Ultrafiltration

- **Coagulation and Flocculation:**
  - Coagulants (e.g., alum, ferric chloride) are added to seawater, which causes suspended particles to aggregate.
  - Flocculation promotes the formation of larger particle clusters (flocs), which are easier to remove.

- **Ultrafiltration:**
    - Membrane-based ultrafiltration units remove colloidal particles, bacteria, and viruses.

Coagulation



Flocculation

Ultrafiltration

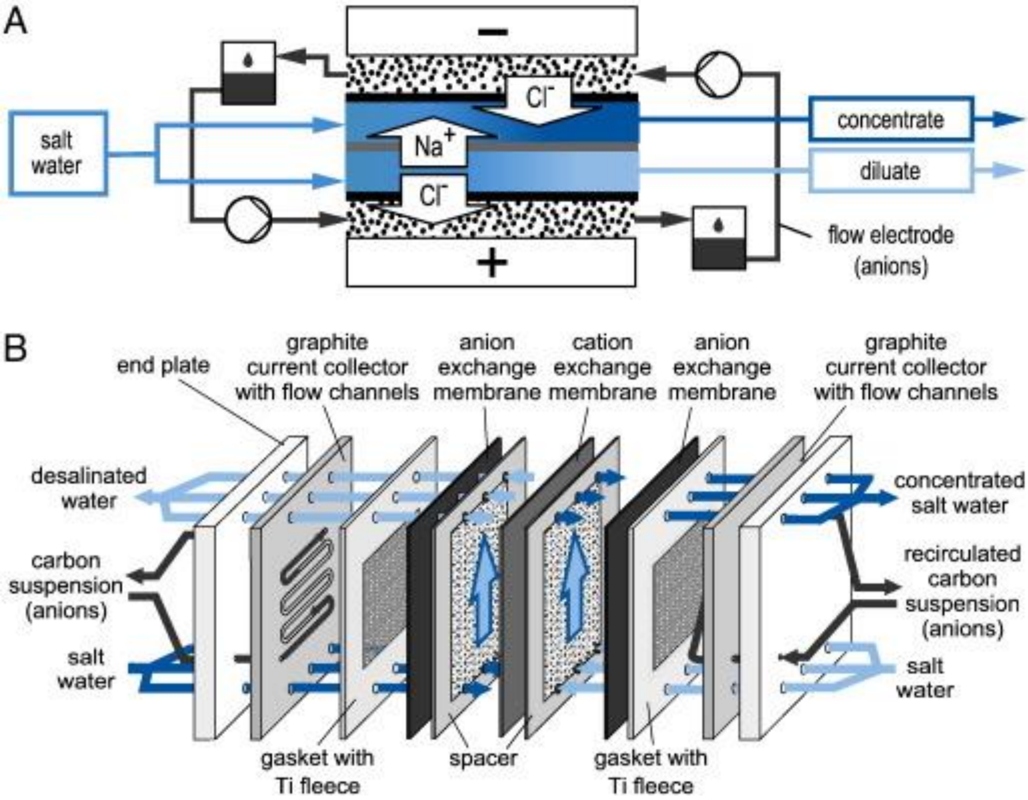
- Pore Size: 0.01 microns, ensuring high-quality water suitable for desalination.
- **Advantages:**
    - Effective removal of both organic and inorganic contaminants.
    - Reduced membrane fouling in the desalination stage.

Desalination Using Forward Osmosis and Deionization

- **Forward Osmosis (FO):**
    - FO is an energy-efficient desalination process that uses a concentrated draw solution to pull water across a semi-permeable membrane, leaving behind salts and other contaminants.
    - The concentrated feedwater, now low in salt content, is then processed through a recovery system, regenerating the draw solution.
        - Oasys Water, USA: FO desalination for industrial brine concentration. Costs: ₹50–80/m³.
        - FO in the Middle East: Pilot plants combining FO with RO to lower overall energy use, achieving costs of ₹40–70/m³.

        - **Material Costs for 200 Tons**:
        - **FO Membranes**: ₹1.85 lakh
        - **Durability**: 5-7 years, subject to fouling and membrane maintenance.
        - **Energy Efficiency:** Compared to traditional reverse osmosis (RO), FO reduces energy c-963 consumption by 30-50%.
- **Deionization:**
    - After forward osmosis, deionization removes residual salts and ions, producing
    - ultrapure water.

- **Technology Used:** Capacitive deionization (CDI), which offers low energy consumption and is scalable.



Material: Ion Exchange Resin (Cation & Anion Resins).

Material Costs for 200 Tons:

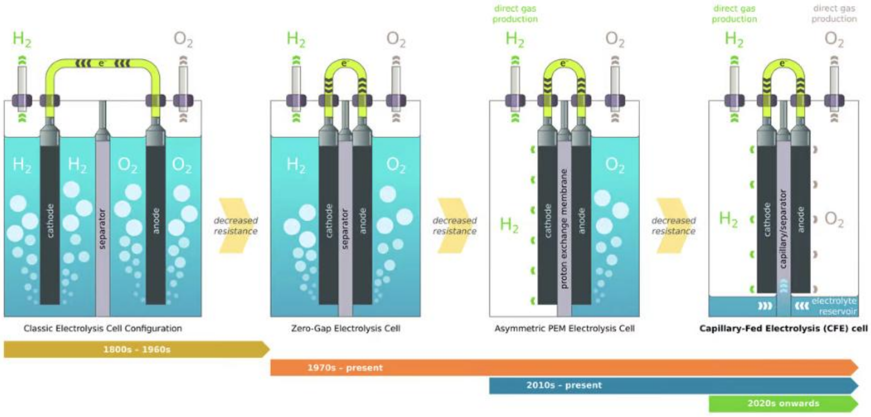
- **Ion Exchange Resin**: ₹1.54 lakh

  Durability: 3-5 years, depending on water quality and resin regeneration cycles.

- **Sizing:** This desalination process will be capable of processing up to more than 30,000 cubic meters of seawater per day.

Hydrogen Production via Capillary-Fed Electrolyser

- **Capillary-Fed Electrolyser Technology:**
  - The purified water from the desalination process enters capillary-fedelectrolysis for hydrogen production.
  - The capillary action ensures uniform water distribution over the electrodes, leading to efficient hydrogen generation.
  - Theelectrolysis are powered by renewable electricity, primarily solar and wind, enabling carbon-free hydrogen production.



- **Electrolyser Specifications:**
  - **Capacity:** 3000 MW electrolyser capacity to meet the 195 TPD hydrogen production.
  - **Conversion Efficiency:** 70-80%, significantly reducing energy losses.
    - **Purpose**: Electrolyzer for hydrogen production with a capillary action system.

Key Materials:

- **Nickel Foam (Anode)**: Provides high surface area for efficient gas evolution.
- **Nickel Mesh (Cathode)**: Essential for conducting hydrogen evolution reactions.
- **Nafion 117 Membrane (Proton Exchange Membrane)**: Ensures ion conductivity for efficient electrolysis.
- **Sintered Metal (Capillary Layer)**: Supports capillary action and enhances the reaction efficiency.
- **Stainless Steel (SS, 10 mm thickness)**: Forms the structural frame, providing mechanical strength.

Material Costs for 195 Tons:

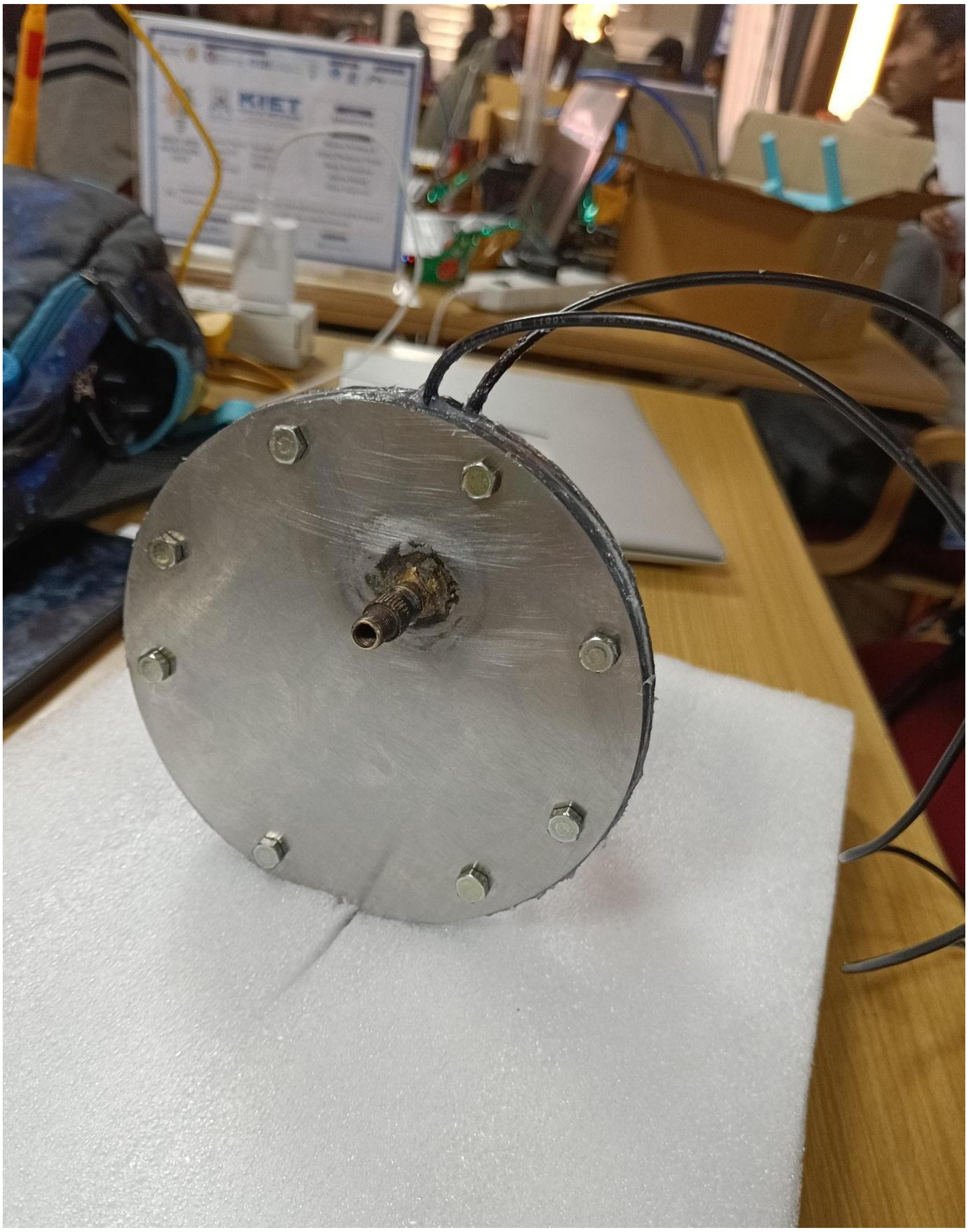
- **Nickel Foam (Anode)**: ₹4.2 lakh
- **Nickel Mesh (Cathode)**: ₹2.3 lakh
- **Nafion 117 Membrane**: ₹61.5 lakh
- **Sintered Metal**: ₹7.7 lakh
- **Stainless Steel (SS)**: ₹4.6 lakh

Total Cost: ₹80.3 lakh

Durability: 5-20 years, depending on material type and operational conditions.

- **Useful Life:** Capillary-fed electrolysis have a useful life of around 20 years, with replacement costs factored into long-term financial planning.

Our prototype



Applications of Green Hydrogen

The hydrogen produced at this facility has a wide range of applications across multiple sectors, contributing to decarbonization and energy transition efforts:

- **Industrial Applications**: Green hydrogen can replace fossil fuels in high-emission industries such as steel manufacturing, cement production, and chemical processing. Hydrogen can serve as a feedstock or energy source, significantly reducing carbon emissions from these sectors.

- **Transportation**: Hydrogen can be used to power fuel cell electric vehicles (FCEVs), particularly in sectors that are difficult to electrify, such as long-distance trucking, aviation, and shipping. FCEVs offer a clean alternative to diesel and gasoline-powered vehicles, helping to reduce transportation-related emissions.
- **Energy Storage**: Green hydrogen can be stored and used to balance the grid during periods of low renewable energy generation. Hydrogen storage offers a long-term energy storage solution, enabling the integration of intermittent renewable energy sources such as solar and wind into the power grid.
- **Power Generation**: Hydrogen can be used in fuel cells or combustion turbines to generate electricity, providing a clean alternative to natural gas. In regions with abundant renewable energy resources, hydrogen power plants can provide reliable, carbon-free electricity.
- **Heating**: Hydrogen can be blended with natural gas or used directly in hydrogen boilers for heating residential and commercial buildings, reducing reliance on fossil fuels for heating applications.

Storage

- **Key Material**: MBU-ABS (Advanced Carbon Composites), known for strength and lightweight properties.
- **Material Costs for 200 Tons**:

- **MBU-ABS Storage**: ₹7.4 crore
- **Durability**: 20+ years, with low maintenance and high strength-to-weight ratio.

Proposed RTC-RE Model

RTC-RE (Round-the-Clock Renewable Energy) Model:

- The plant is designed to operate 24/7 using renewable energy. Solar energy will provide daytime electricity, while wind energy, along with energy storage systems, will maintain operations during non-daylight hours.
- **Energy Storage:** Battery systems will store excess solar energy for use during the night, ensuring uninterrupted hydrogen production.

Proposed Sourcing of Feedstock

- **Seawater Intake:** The facility will source seawater directly from the coast, with a daily intake of around 35,000 cubic meters. The intake structure will be designed to minimize environmental impact and prevent marine life entrapment.
- **Renewable Energy:** The plant will rely on solar and wind energy installations, potentially through power purchase agreements (PPAs) with renewable energy providers or captive installations.

Plant Capacity Utilization Factor (CUF)

- **CUF Estimate:** 85-90%, based on the availability of renewable energy and system maintenance.
- High CUF ensures efficient use of assets and maximizes hydrogen production year-round.

Financial model

The financial model includes capital expenditures (CAPEX), operating expenses (OPEX), and projected revenues from the sale of hydrogen (H_2) and oxygen (O_2) as by-products.

3. **Phased Approach for Hydrogen Production**

- **Phase 1:** 65 tons/day
- **Phase 2:** Additional 65 tons/day
- **Phase 3:** Additional 50 tons/day (for a total of 180 tons/day)

3.3. Production Capacity Breakdown

- **Phase 1:** 65 tons/day × 365 days = 23,725 tons/year
- **Phase 2 (Cumulative):** 130 tons/day × 365 days = 47,450 tons/year
- **Phase 3 (Final):** 195 tons/day × 365 days = 71175 tons/year

2. Revised Financial Model

The financial model reflects the production of 65 tons per day in the initial phases, scaling up to 195 tons per day upon completion of the final phase.

2.1. CAPEX Breakdown (₹1500 crore Total)

Phase 1 (65 TPD): ₹500 crore

- **Seawater Intake and Primary Treatment:** ₹65 crore
- **Desalination (FO and CDI):** ₹100 crore
- **Hydrogen Production (Electrolysers):** ₹200 crore
- **Renewable Energy Systems (Solar, Wind, Storage):** ₹100 crore
- **Other Infrastructure and Contingencies:** ₹35 crore

Phase 2 (Additional 65 TPD): ₹500 crore

- **Seawater Intake and Treatment Expansion:** ₹50 crore
- **Desalination:** ₹100 crore
- **Hydrogen Production (Electrolysers):** ₹200 crore
- **Renewable Energy Expansion:** ₹100 crore
- **Other Infrastructure:** ₹50 crore

Phase 3 (Additional 50 TPD, Final 180 TPD): ₹500 crore

- **Seawater Intake Optimization:** ₹40 crore
- **Desalination:** ₹100 crore
- **Hydrogen Production (Electrolysers):** ₹200 crore
- **Renewable Energy Expansion:** ₹100 crore
- **Miscellaneous Infrastructure:** ₹60 crore

Total CAPEX for 180 TPD plant (Phased): ₹1500 crore

2.2. OPEX Breakdown (₹65 crore per phase, ₹195 crore total per annum)

- **Energy Costs (Renewables):** ₹25 crore/phase (₹75 crore total at full capacity)
- **Maintenance (Electrolysers, Desalination, Storage):** ₹12 crore/phase (₹36 crore total)
- **Feedstock (Seawater, Chemicals):** ₹8 crore/phase (₹24 crore total)
- **Labor and Operations:** ₹20 crore/phase (₹60 crore total)

Total OPEX:

- **Phase 1 OPEX:** ₹65 crore/year
- **Phase 2 OPEX:** ₹130 crore/year (Cumulative)
- **Phase 3 OPEX:** ₹195 crore/year (Full Capacity)

3. Revenue Generation from Hydrogen (H_2) and Oxygen (O_2)

The revenue from hydrogen is based on the revised pricing in each phase. Oxygen revenue remains constant.

3.1. Hydrogen Sales Revenue

Phase 1 (65 TPD, ₹99/kg):

- **Daily Hydrogen Production:** 65,000 kg/day
- **Revenue from H_2 (₹99/kg):** ₹65,000 × 99 = ₹64,35,000 lakh/day
- **Annual H_2 Revenue:** ₹64,35,000 lakh/day × 365 = ₹234,87,75000 crore/year

Phase 2 (130 TPD, ₹90/kg):

- **Daily Hydrogen Production:** 130,000 kg/day
- **Revenue from H_2 (₹90/kg):** ₹130,000 × 90 = 11700000 lakh/day
- **Annual H_2 Revenue:** ₹11700000 lakh/day × 365 = ₹427,05,00,000 crore/year

Phase 3 (180 TPD, ₹83/kg):

- **Daily Hydrogen Production:** 195,000 kg/day
- **Revenue from H_2 (₹83/kg):** ₹195,000 × 83 = ₹1,61,85000 lakh/day
- **Annual H_2 Revenue:** ₹1,61,85000 lakh/day × 365 = ₹590,75,25000crore/year

3.2. Oxygen By-Product Revenue

Oxygen revenue is calculated at ₹16/kg across all phases.

Phase 1 (65 TPD):

- **Oxygen Production:** 16,250 kg/day
- **Revenue from O_2 (₹15/kg):** ₹16,250 × 15 = ₹2,43,750 lakh/day
- **Annual O_2 Revenue:** ₹2,43,750 lakh/day × 365 = ₹8,89,98,750crore/year

Phase 2 (130 TPD)

- **Oxygen Production:** 32,500 kg/day
- **Revenue from O_2 (₹15/kg):** ₹32,500 × 15 = ₹4,87,500lakh/day
- **Annual O_2 Revenue:** ₹4,87,500 lakh/day × 365 = ₹17,79,37,500 crore/year

Phase 3 (180 TPD):

- **Oxygen Production:** 48,750 kg/day
- **Revenue from O_2 (₹15/kg):** ₹48,750 × 15 = ₹7,31,250 lakh/day
- **Annual O_2 Revenue:** ₹7,31,250 lakh/day × 365 = ₹26,69,06,250 crore/year

3.3. Total Revenue (Hydrogen + Oxygen)

Phase 1 (65 TPD):

- **Total Daily Revenue:** ₹64,35,000 lakh/day`f (H_2) + ₹2,43,750 lakh/day (O_2)
- **Annual Revenue (H_2 + O_2):** ₹66,78,750 lakh/day × 365 = ₹243,77,43,750 crore/year

Phase 2 (130 TPD):

- **Total Daily Revenue:** ₹1,17,00,000 lakh (H_2) + ₹4,87,500 lakh (O_2)
- **Annual Revenue (H_2 + O_2):** ₹1,21,87,500 lakh/day × 365 = ₹4,44,84,37,500 crore/year

Phase 3 (180 TPD):

- **Total Daily Revenue:** ₹1,61,85,000 lakh (H_2) + ₹7,31,250 lakh (O_2)
- **Annual Revenue (H_2 + O_2):** ₹1,69,16,250 lakh/day × 365 = ₹6,17,44,31,250 crore/year

4. Profitability and Payback Period

4.1. Profit Margin (Phase 3 - Full Production)

- **Annual Revenue (Full Production):** ₹617.44 crore/year
- **Annual OPEX (Full Production):** ₹200 crore/year
- **Annual Profit (Before Interest, Depreciation, and Tax):** ₹617.44 crore – ₹200 crore = 417.66 crore/year

4.2. Payback Period

- **Total CAPEX:** ₹1500 crore
- **Annual Profit (After Full Capacity):** ₹417.66 crore/year
- **Payback Period:** ₹1500 crore / ₹417.44 crore ≈ 4 years

Approaches to Lower Levelized Cost of Hydrogen (LCOH)

The LCOH decreases across phases due to improved economies of scale:

- **Phase 1:** Initial LCOH is higher at ₹99/kg.
- **Phase 2:** Cost reductions and efficiencies lead to a hydrogen price of ₹90/kg.
- **Phase 3:** Full-scale production achieves a hydrogen price of ₹83/kg.

Strategies to reduce LCOH:

- **Phased Development:** Revenue from early phases helps finance later phases.
- **Renewable Energy Utilization:** Efficient use of solar and wind energy.

- **Technological Advancements:** Improved electrolyser and desalination technologies.

Hydrogen Storage Technologies

- **Modular Composite Tanks:** Hydrogen will be stored in carbon Fiber composite tanks operating at lower pressures (200-350 bar), offering both safety and cost benefits.
- **Storage Cost:** The cost of carbon Fiber tanks is projected to reduce by 10-20% over time as composite material technologies improve.

PRESENT HYDROGEN MARKET IN INDIA, CHINA, AND USA EUROPE

PRESENT HYDROGEN MARKET IN INDIA:

Hydrogen production technology in India is gaining traction due to its potential to support the transition to clean energy. The country is focusing on scaling up hydrogen production to meet its increasing energy demands and reduce dependence on fossil fuels.

1. Current Hydrogen Production Technologies in India

a. Grey Hydrogen

- **Source:** Produced from natural gas or coal using steam methane reforming (SMR) or coal gasification.
- **Emissions:** Significant CO_2 emissions.
- **Market Presence:** Predominant method due to lower costs and established infrastructure.
- **Challenges:** Environmental concerns and carbon footprint.

b. Blue Hydrogen

- **Source:** Similar to grey hydrogen but with carbon capture and storage (CCS) to reduce emissions.
- **Market Presence:** Emerging in India as CCS technology develops.
- **Opportunities:** Potential to retrofit existing grey hydrogen plants.
- **Challenges:** High cost of CCS and lack of infrastructure.

c. Green Hydrogen

- **Source:** Produced using electrolysis powered by renewable energy (solar, wind, hydro).
- **Market Presence:** Still nascent but rapidly gaining momentum due to policy push.
- **Advantages:** Zero emissions during production.
- **Challenges:** High production cost and limited renewable energy infrastructure.

2. Current Market Landscape in India

a. Demand Sectors:

Ammonia and Fertilizer Industry:

- Largest consumer of hydrogen in India.

- Shift towards green hydrogen for sustainable ammonia production.

Petroleum Refining:

- Used for desulfurization and hydrocracking.
- Efforts to decarbonize this sector are underway.

Steel and Cement:

- Pilot projects exploring hydrogen as a replacement for coking coal.

Transportation:

- Hydrogen fuel cell vehicles (FCEVs) are gaining attention for long-haul transportation.
- Focus on hydrogen buses and trucks in the early adoption stage.

Power Generation:

- Green hydrogen is being tested for blending with natural gas in power plants.

b. Policy Initiatives

- **National Green Hydrogen Mission (2023):**
    - Target of 5 million metric tonnes of green hydrogen production annually by 2030.
    - Incentives for production, R&D, and infrastructure development.
- **Renewable Energy Integration:**
    - Focus on leveraging India's vast solar and wind potential to power electrolysis.

3. Emerging Technologies and Trends

a. Advanced Electrolysis

- **PEM (Proton Exchange Membrane) Electrolysis:** Efficient and adaptable for renewable energy integration.
- **AEM (Anion Exchange Membrane) Electrolysis:** Cost-effective alternative with emerging R&D focus.

b. Biomass Gasification

- Potential for decentralized hydrogen production using agricultural waste.

c. Forward Osmosis (FO) and Capacitive Deionization (CDI)

- Your project aligns with the innovative approach of integrating FO and CDI for desalination and low-energy water preparation for electrolysis.

d. Capillary-fed Electrolyzers

- Cutting-edge technology for efficient water splitting, reducing costs and enhancing hydrogen yield.

4. Challenges and Opportunities:

Challenges

- High cost of green hydrogen production (~$3-6/kg).

- Limited hydrogen storage and distribution infrastructure.
- Need for technology localization and indigenization.

Opportunities

- Abundant renewable energy resources for green hydrogen.
- Growing interest in hydrogen blending in industrial and domestic pipelines.
- Potential for hydrogen exports, especially to Europe and East Asia.

Development

The hydrogen market in India is rapidly expanding, with a strong focus on green hydrogen as a cornerstone of its energy transition strategy. India aims to position itself as a global hub for green hydrogen production, utilization, and export, supported by ambitious government initiatives and private sector investments.

Key Developments:

1. **National Green Hydrogen Mission**:
    - The mission, with an investment of $2.4 billion, aims to achieve an annual green hydrogen production capacity of 5 million metric tons (MMT) by 2030. This includes significant investment in renewable energy to power electrolyzers and promote hydrogen as a fuel for industrial and transportation sectors.
2. **Infrastructure and Pilot Projects**:
    - Several public sector undertakings (e.g., NTPC, Indian Oil) are deploying hydrogen blending projects in natural gas grids and hydrogen fuel cell buses. This includes blending green hydrogen into city gas grids and the development of fuel cell electric vehicles (FCEVs).
3. **Market Size and Future Potential**:
    - The Indian hydrogen market was valued at approximately $50 billion in 2022 and is projected to grow significantly, driven by demand in refining, fertilizers, and steel production. Key players are investing in large-scale projects to reduce dependency on imported energy and facilitate export of green hydrogen derivatives.
4. **Global Leadership in Project Announcements**:
    - In early 2024, India accounted for 58% of global green hydrogen project announcements. This reflects a transition from policy frameworks to actionable implementation.

Profitability in India's Hydrogen Market

Companies Profiting in the Hydrogen Sector

1. **Major Public Sector Undertakings (PSUs):**
    - **Indian Oil Corporation (IOC)**: Profits from hydrogen blending and pilot green hydrogen projects in refineries. They also invest in solar-powered electrolyzers and hydrogen fuel refueling infrastructure.

- **NTPC Limited**: Profits through hydrogen pilot projects for fuel cells and power generation. NTPC is leveraging renewable energy to produce green hydrogen.
- **GAIL India**: Recently initiated hydrogen blending in city gas distribution, exploring cost-effective solutions and tapping into established natural gas networks.

2. **Private Companies:**
   - **Reliance Industries**: Large investments in green hydrogen production and technology under their clean energy business. Their focus on cost-efficient electrolysis technology provides a competitive edge.
   - **Adani Group**: Plans to produce green hydrogen at scale and integrate it into industrial applications such as fertilizers and steel.
   - **ReNew Power**: Engaged in green hydrogen projects with a focus on renewable energy integration for scalable hydrogen production.
3. **Startups and Innovators:**
   - Emerging startups focus on modular electrolyzers, hydrogen storage solutions, and innovative transport mechanisms, benefitting from government subsidies and private investments.

Types of Competitors

1. **Sector-based Competitors**:
   - **Fertilizer Industry**: Companies like Rashtriya Chemicals & Fertilizers and Tata Chemicals compete in adopting green hydrogen for ammonia production.
   - **Steel Producers**: Tata Steel and JSW Steel are piloting hydrogen use to decarbonize steel production.
   - **Oil and Gas**: BPCL and HPCL are exploring hydrogen blending to maintain competitiveness against other energy solutions.
2. **Technology Competitors**:
   - **Electrolyzer Manufacturers**: Global players such as ITM Power and local companies like Ohmium and Reliance are vying to produce cost-efficient electrolyzers in India.
   - **Hydrogen Storage and Transportation**: Firms innovating in hydrogen storage, like Shell and Linde, compete with domestic startups developing cost-effective alternatives.
3. **Renewable Energy Integration**:
   - **Solar and Wind Producers**: Companies like ReNew Power and Suzlon compete to provide renewable energy for green hydrogen projects, ensuring their position as suppliers in the value chain.

4. **International Competition**:

- European and Middle Eastern hydrogen producers, such as Air Liquide and Saudi Aramco, aim to capture a share of India's market or compete in exports.

Projected Growth Trends (2024-2030)

Green Hydrogen Production Capacity:

Target: 5 million metric tonnes (MMT) of green hydrogen annually by 2030 under the National Green Hydrogen Mission.

Investment: Estimated $2.4 billion allocated for the development of hydrogen production infrastructure and research.

Market Value:

The hydrogen market is expected to grow from ~$50 billion in 2022 to over $90 billion by 2030, propelled by demand in refining, steel, and fertilizer sectors.

Renewable Energy Integration:

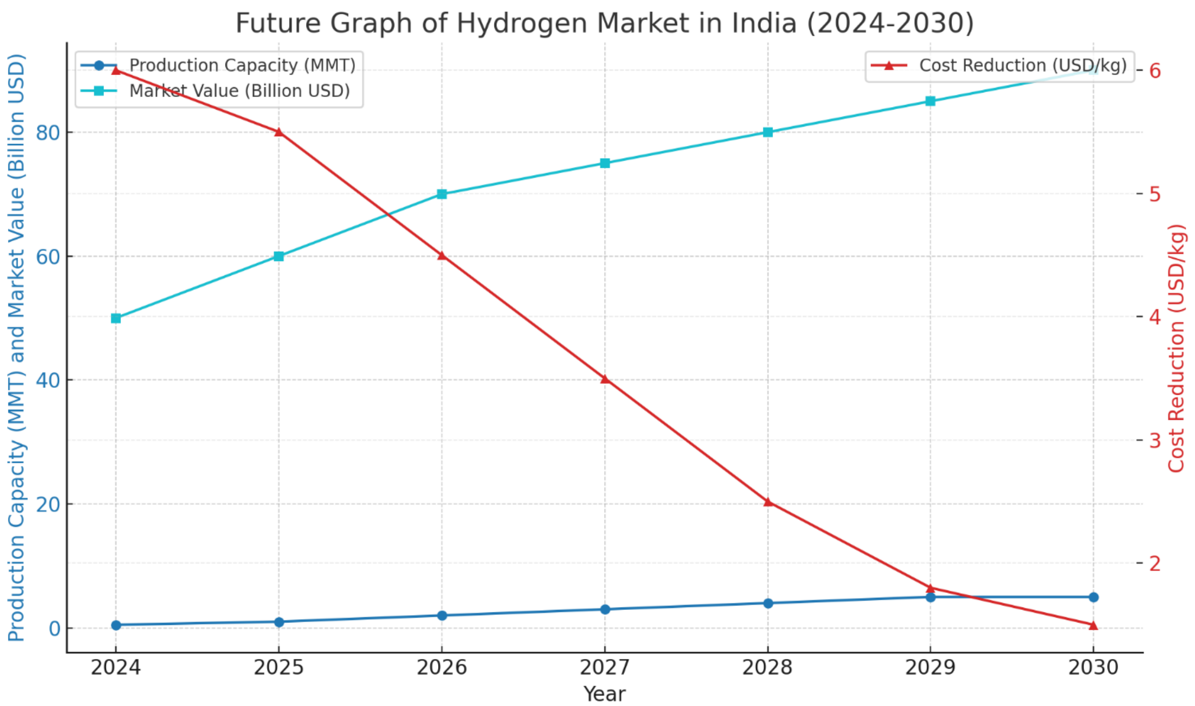
India aims to dedicate 125 GW of renewable energy capacity specifically for green hydrogen production, leveraging its vast solar and wind resources.

Export Potential:

India has the potential to become a global hub for green hydrogen exports, particularly to Europe and East Asia, with significant investments in port infrastructure and export facilities.

Cost Reduction:

Green hydrogen production costs, currently at $3-6/kg, are expected to decrease to $1-2/kg by 2030, thanks to advancements in electrolyzer technology and renewable energy efficiency.



Here is the graph depicting the future of India's hydrogen market (2024-2030):

Blue Lines:

- Production Capacity (in MMT): Shows an exponential rise to meet the 5 MMT target by 2030.
- Market Value (in billion USD): Indicates steady growth driven by increasing adoption and investment.

Red Line:

- Cost Reduction (in USD/kg): Represents a significant decline, making green hydrogen more competitive.

PRESENT HYDROGEN MARKET IN CHINA:

1. Current Hydrogen Production Technologies

a. Coal Gasification (Predominant)

- Market Share: ~60% of hydrogen production in China.
- Process: Coal is a gasifier with oxygen and steam to produce synthesis gas (H_2 + CO).

Advantages:

- Abundant coal reserves in China.
- Cost-effective due to low coal prices.

Challenges:

- High carbon emissions.
- Not sustainable in the long term without Carbon Capture and Storage (CCS).

b. Steam Methane Reforming (SMR)

- Market Share: ~20-25%.
- Process: Natural gas reacts with steam under high temperature to produce hydrogen and CO_2.

Advantages:

- Efficient and mature technology.
- Lower emissions than coal gasification.

Challenges:

- Dependence on imported natural gas.
- CO_2 emissions still significant.

c. Electrolysis (Emerging)

- Market Share: ~3-5%, but growing.
- Process: Splits water into hydrogen and oxygen using electricity.

Advantages:

- Green hydrogen if powered by renewable energy (solar, wind, or hydro).
- No direct CO_2 emissions.

Challenges:

- High costs due to electricity expenses.
- Efficiency improvements needed.

d. Other Methods (Minor Contributions)

- Biomass Gasification: Hydrogen from agricultural or forest waste.
- By-product Hydrogen: Generated from industries such as chlor-alkali production or steelmaking.

2. Market Trends and Future Directions

a. Shift Towards Green Hydrogen

- **Policies:** Government incentives and subsidies aim to expand renewable energy-driven hydrogen production.
- **Target:** Increase the share of green hydrogen significantly by 2035.

b. Infrastructure Development

Hydrogen Hubs: China is building hydrogen clusters in regions like Guangdong, Hebei, and Sichuan.

Fueling Stations: Over 300 hydrogen refueling stations (HRS) operational, the highest globally.

c. Research and Innovation

Focus on improving the efficiency and scalability of emerging technologies like:

- Solid Oxide Electrolysis Cells (SOECs)
- Photocatalytic Water Splitting
- Thermochemical Splitting

d. Role of Hydrogen in Transportation and Power

- Fuel Cell Vehicles (FCVs): Over 12,000 FCVs on Chinese roads as of 2023.
- Energy Storage: Hydrogen as a long-term storage solution for intermittent renewable energy.

Challenges in China's Hydrogen Market

Cost Competitiveness: Green hydrogen is still more expensive than blue or Gray hydrogen.

Supply Chain Development: Transport and storage of hydrogen require specialized infrastructure.

Regulatory Framework: Need for standardized policies to ensure market stability

Market Growth of China :
s
China's hydrogen market is experiencing significant growth and development, driven by strong government policies, technological advancements, and the expansion of renewable energy capacity.

1. Market Size and Growth:
    - The hydrogen energy industry in China is expected to reach an output value of 1 trillion yuan (approximately $14 billion) by 2025. By 2030, the annual production of

hydrogen could scale up to 43 million tonnes, with green hydrogen making up 10% of the energy mix compared to just 1% in 2019.

- The industrial value chain is projected to grow to 12 trillion yuan by 2025, showing the comprehensive scale of planned investments and infrastructure.

2. Production and Infrastructure:

    - China installed over 1 GW of electrolyzer capacity in 2023, a global high, but a large portion of current hydrogen production still comes from grey hydrogen methods like coal gasification and steam methane reforming. Transitioning to green hydrogen is essential for achieving China's carbon neutrality goals by 2060.
    - The country leads the world in hydrogen refueling stations, with over 400 built and 280 operational. Key provinces like Inner Mongolia and Gansu aim to significantly increase green hydrogen production by leveraging abundant renewable energy sources.

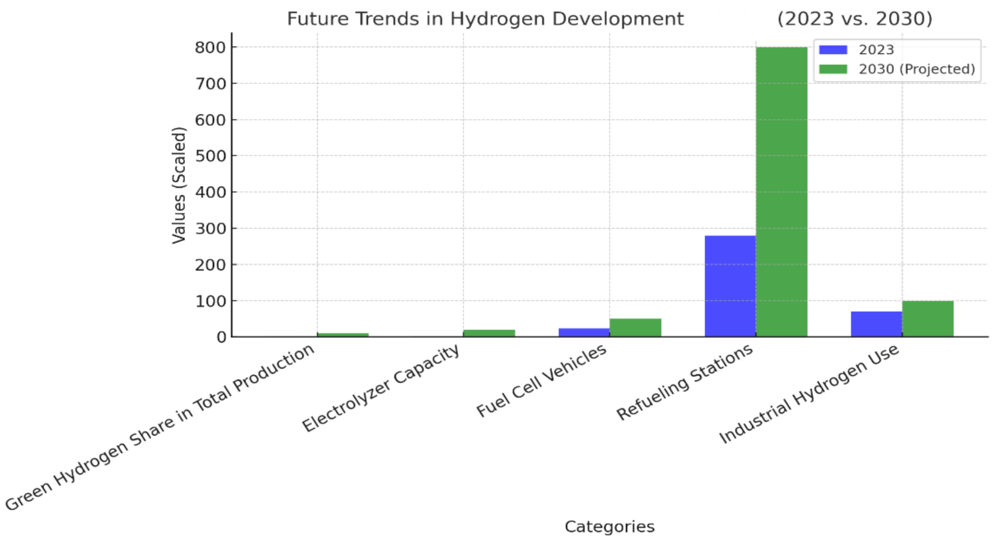
3. Applications:

    - The transportation sector is a key focus area. The number of hydrogen fuel-cell vehicles in China is projected to grow from around 22,790 currently to 50,000 by 2025. Infrastructure and vehicle manufacturing incentives support this.
    - Hydrogen is also being used in industrial applications, power generation, and as a clean fuel for various processes, with advancements spanning production, storage, and transportation technologies.

4. Challenges and Innovations:

    - China faces challenges like geographical mismatches between renewable energy sources (north and west) and hydrogen demand centres (east). Projects like Sinopec's hydrogen pipelines aim to address this by creating long-distance conduits.
    - Renewable energy-based hydrogen projects are expanding rapidly. For example, Inner Mongolia and Xinjiang are focusing on green hydrogen with targets of 480,000 tonnes and 200,000 tonnes annually by 2025.

Future Growth Of Hydrogen Gas In India



The diagram above illustrates the projected trends in hydrogen development in China, comparing the current state (2023) to projections for 2030 across key categories:

1. **Green Hydrogen Share in Total Production**: Expected to grow significantly from 1% in 2023 to 10% by 2030.
2. **Electrolyzer Capacity**: Projected to increase from 1 GW to 20 GW.
3. **Fuel Cell Vehicles**: Anticipated to grow from 22,800 to 50,000 units.
4. **Hydrogen Refueling Stations**: Expansion from 280 stations in 2023 to around 800 by 2030.
5. **Industrial Hydrogen Use**: Projected increase reflecting broader adoption across sectors.

This growth is driven by advancements in technology, infrastructure development, and supportive government policies.

Impact of Hydrogen Gas Production in India

The rise of hydrogen gas production in India has transformative implications across economic, environmental, and industrial dimensions. Here's an overview:

1. Economic Impact

- **Energy Security**:
    - Hydrogen reduces dependency on imported fossil fuels, saving billions annually (India imports ~$160 billion worth of crude oil each year).
- **Job Creation**:
    - Green hydrogen projects are expected to generate thousands of jobs in production, distribution, and infrastructure development.
- **Export Revenue**:
    - By positioning itself as a global hub, India could earn significant revenue from green hydrogen exports to regions like Europe and East Asia.

2. Environmental Impact

- **Carbon Emission Reduction**:
    - Transitioning from grey to green hydrogen can help India achieve its net-zero goal by 2070.
    - Sectors like steel, cement, and transportation will drastically lower CO_2 emissions using green hydrogen.
- **Air Quality Improvement**:
    - Hydrogen adoption in transportation reduces particulate emissions, improving urban air quality.

3. Industrial Transformation

- **Decarbonizing Heavy Industries**:

- Industries like steel and cement, which traditionally rely on coal, will adopt hydrogen as a clean fuel.

- **Ammonia and Fertilizer Production**:
    - Green hydrogen enables sustainable production of ammonia, reducing reliance on natural gas-based processes.
- Hydrogen Blending in Energy Systems:
    - Blending hydrogen with natural gas will improve energy system efficiency and lower emissions.

Vision for Hydrogen Gas Production in India

India's vision for hydrogen production aligns with its broader goals of energy transition, industrial competitiveness, and environmental sustainability:

1. Energy Independence

- Develop a robust domestic hydrogen ecosystem powered by abundant renewable energy resources.
- Achieve a production target of **5 million metric tons (MMT) annually** by 2030 under the **National Green Hydrogen Mission**.

2. Technological Leadership

- Foster innovation in electrolyzers, hydrogen storage, and transportation technologies.
- Invest in R&D to indigenize technology, reducing reliance on imports.

3. Global Export Hub

- Become a key exporter of green hydrogen and its derivatives (e.g., green ammonia and methanol), leveraging low-cost production and strategic trade agreements.

5. Inclusive and Sustainable Growth

- Encourage participation of startups and SMEs in the hydrogen value chain.
- Ensure sustainability by integrating hydrogen production with renewable energy and water recycling technologies.

Policy and Regulatory Support by the Indian Government

The Indian government has been actively supporting the development and adoption of green hydrogen through various policies and regulatory measures. Here are some key initiatives and support mechanisms:

1. National Hydrogen Mission

- **Launch and Objectives:** The National Hydrogen Mission was launched in 2021 with the aim to make India a global hub for green hydrogen production and export. The mission focuses on generating hydrogen from renewable energy sources to reduce carbon emissions and dependence on fossil fuels.
- **Incentives and Subsidies:** The government provides financial incentives and subsidies to promote research and development, pilot projects, and large-scale deployment of green hydrogen technologies.

2. Policy Framework

- **Renewable Purchase Obligations (RPOs):** The government mandates certain percentages of energy consumption to be met from renewable sources, including green hydrogen, to encourage its adoption.

- **Green Hydrogen Consumption Obligations (GHCOs):** Similar to RPOs, GHCOs require industries to use a certain percentage of green hydrogen in their processes, particularly in sectors like refining and fertilizer production.

3. Financial Support and Funding

- **Viability Gap Funding (VGF):** To make green hydrogen projects financially viable, the government offers VGF to cover the gap between the cost of production and the market price.
- **Tax Benefits and Exemptions:** Various tax incentives, such as reduced GST rates and income tax exemptions, are provided to companies involved in green hydrogen production and infrastructure development.

4. Infrastructure Development

- **Hydrogen Corridors:** The government is planning to develop hydrogen corridors to facilitate the transportation and distribution of green hydrogen across the country.
- **Refueling Stations:** Support for the establishment of hydrogen refueling stations to promote the use of hydrogen fuel cell vehicles.

5. Research and Development

- **Collaborations and Partnerships:** The government encourages collaborations between public and private sectors, as well as international partnerships, to advance green hydrogen technologies.
- **Funding for Innovation:** Grants and funding are provided for innovative projects and startups working on green hydrogen solutions.

6. Regulatory Measures

- **Safety Standards and Certifications:** The government has established safety standards and certification processes to ensure the safe production, storage, and transportation of hydrogen.
- **Environmental Regulations:** Policies are in place to ensure that green hydrogen production adheres to environmental regulations and contributes to sustainability goals.

7. Global Engagement

- **International Cooperation:** India is actively participating in international forums and collaborations to promote the global adoption of green hydrogen and to attract foreign investments in the sector.
- **Export Opportunities:** The government is exploring opportunities to export green hydrogen to countries with high demand, positioning India as a key player in the global hydrogen market.

These initiatives reflect the Indian government's commitment to fostering a robust green hydrogen ecosystem, which is crucial for achieving the country's climate goals and transitioning to a sustainable energy future

Conclusion

Green hydrogen is set to play a crucial role in India's energy transition, especially in the decarbonization of hard-to-abate sectors. The National Green Hydrogen Mission is a significant step in this direction. This mission aims to facilitate the development of a green hydrogen ecosystem and create opportunities for innovation and investment throughout the green hydrogen value chain. This approach is expected to lead to increased investments, job creation, and economic growth. Government interventions will kickstart this process and provide the necessary support to unlock market potential across various sectors by promoting cost reduction and economies of scale.

Research and references

- **Hydrogen Production from Seawater Electrolysis: Current Status and Future Prospects** by J. A. Turner
- **Hydrogen Production: Technologies and Applications** by Mehmet Sankir and Nurdan Demirci Sankir
- **Forward Osmosis: Principles, Applications, and Recent Developments** by Menachem Elimelech and William A. Phillip https://www.researchgate.net/publication/299141670_Water_desalination_via_capacitive_deionization
- **International Renewable Energy Agency (IRENA)**: IRENA
- **A high-performance capillary-fed electrolysis cell promises more cost-competitive renewable hydrogen** by Aaron Hodges1,3, Anh Linh Hoang1,3, George Tsekouras1 ,
- Klaudia Wagner1,2, Chong-Yong Lee 1,2, Gerhard F. Swiegers 1,2 & Gordon G. Wallace 1,2 **https://www.researchgate.net/publication/369203038_An_Overview_of_the_Recent_Advances_in_Composite_Materials_and_Artificial_Intelligence_for_Hydrogen_Storage_Vessels_Design**
- “National Green Hydrogen Mission,” Government of India, January 4, 2022, https://www.india.gov.in/spotlight/national-green-hydrogen-mission
- 1. Hydrogen Market in India
  - **Author**: Gurpreet Chugh
  - **Source**: Asia Clean Energy Forum
  - **Summary**: This presentation discusses India's hydrogen demand projections, potential market size, and the role of renewable energy in hydrogen production.
  - https://asiacleanenergyforum.adb.org/wp-content/uploads/2023/06/Gurpreet-Chugh.pdf

- 2. Hydrogen Market Report
  - **Publisher**: Federation of Indian Petroleum Industry (FIPI)
  - **Summary:** This report examines hydrogen demand in India, green hydrogen production technologies, cost economics, storage and transportation challenges, and policy recommendations.
  - https://www.fipi.org.in/assets/pdf/downstream-report/Hydrogen%20Market%20Report.pdf

- Green Hydrogen Demand Assessment for Commercial & Industrial Consumers in India
  - **Publisher:** World Wide Fund for Nature (WWF) India
  - **Summary**: This report analyzes the current hydrogen market in India, focusing on key industries, demand assessment, and potential adoption of green hydrogen by commercial and industrial consumers.
  - https://wwfin.awsassets.panda.org/downloads/green-hydrogen-demand-assessment.pdf
- Decoding India's Green Hydrogen Potential
  - **Publisher**: Observer Research Foundation (ORF) America
  - **Summary:** This research paper evaluates India's green hydrogen production costs, potential for scaling up, and the role of policy in fostering a green hydrogen economy.
  - https://orfamerica.org/newresearch/green-hydrogen-bp